Effect of Image Enhancement in CNN-Based Medical Image Classification: A Systematic Literature Review

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***Abstract*—Diagnosing diseases is one of the most crucial tasks in the medical field. One of the approaches commonly used is by analyzing medical images. However, detecting diseases manually through images may take considerable time and effort. Therefore, many researchers have developed image classification methods for faster disease recognition. Nevertheless, several problems, such as low-quality images, may occur during the process. It is suggested that applying image enhancement to medical scans may improve image classification performance. Despite this, the effects of image enhancement itself on medical image classification performance have not been discussed deeply. To fill the gap, this study provides a systematic literature review of how applying image enhancement during the preprocessing step affects classification performance, along with algorithms and approaches used in image enhancement and classification processes. In most cases, applying image enhancement to the images improves medical image classification performance. Upon further research, all algorithms listed in this systematic literature review has their own characteristic that may positively impact medical image classification performance, depending on the dataset used. Several researchers have also attempted to combine multiple image enhancements to complement their disadvantages. However, not all attempts managed to improve the classification performance. Some situations where the dataset is unsuitable with the image enhancement algorithm have shown worse performance than the respective original image.**

Keywords—Image Preprocessing, Image Enhancement, Classification, CNNs, Medical Images

I. Introduction

Image classification is a process of categorizing images into different classes based on certain rules. Image Classification is needed in issues of various areas, such as medical [1], autonomous vehicle [2], face recognition [3], etc. To address these problems, many techniques have been developed by practitioners and researchers throughout the decade. One of them is the Convolutional Neural Network (CNN), which has been used extensively in different image classification applications [4]. Despite this, most publications fail to address problems in their dataset quality.

Image enhancement is a method of improving the quality of the information in images for better interpretability on human viewers and more refined input for automated image processing techniques [5]. Some of the methods involved are intensity and contrast manipulation, noise reduction, edge sharpening, and filtering [6]. In image classification, the purpose of image enhancement is to highlight certain image features. Therefore, applying image enhancement during the preprocessing step may affect image classification performance. Although the process is frequently used in many proposed image classification techniques, its effect towards the classification performance has not been explored in its detail. Rahman et al. [7] suggest that using Contrast Limited Adaptive Histogram Equalization (CLAHE) improves the accuracy of proposed COVID-19 chest X-ray (CXR) images classification by 0.62%. However, Friere et al. [8] stated that CLAHE does not increase the proposed performance but decreases the precision by 0.10% compared to the original data. Thus, it is hard to conclude how image enhancement affect image classification in general, especially in healthcare.

Therefore, this paper aims to assess the effectiveness of applying image enhancement(s) in the medical image dataset before inserting said dataset into a CNN-Based classifier. We hope to clarify how this approach affects classification performance and find the best method for some medical image classification cases. The paper gives an overview of efforts made by researchers regarding the issue, including its results, merits, and demerits. The paper is structured as follows. Section II discussed the definition of image classification and image enhancement, and how researchers have incorporated them to solve medical image classification problems. Section III introduces the methodology of the SLR. Section IV presents the literature retrieved and analyzes them based on the research questions formulated in Section III. Lastly, Section V concludes the SLR with a summary.

II. Background

A. Image Classification and Its Approaches

Image classification is the task of categorizing images into one of the predetermined classes. It is one of the fundamental tasks in computer vision [9]. In general, the process of image classification involves the selection of training samples, image preprocessing, feature extraction, selection of classification

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approaches, post-classification processing, and performance evaluation [10]. Throughout time, various classification approaches have been developed and applied. Among these approaches, Lecun et al. [11] stated that the method dominantly used for image classification, recognition, and detection is CNN.

As one of the major process in image classification, image preprocessing is a crucial step yet often overlooked [6]. In real life, image quality may vary due to external factors, leading to worse performance. One of the examples is X-ray images taken in the hospital for COVID-19 patients, where the quality of images may vary due to multiple reasons: condition of the patients, breathing state, and human error [7]. Therefore, image preprocessing must be done before proceeding to the next step.

B. Image Enhancement Techniques

Image enhancement is a method that aims in improving the image to be more suitable for some applications by reducing or removing certain secondary information [7]. The need for image enhancement varies on how the images were taken. In some cases, multiple image enhancement algorithms must be applied to improve the image better than the original one. Abu Samah et al. [12] use both high pass filter (HF) and histogram equalization (HE) as only applying the HF will enhance the image's high-intensity gradient but significantly affect the color intensities. So, HE must be involved to improve the visual. Therefore, it is essential to know which image enhancement methods or algorithms need to be used.

C. Image Enhancement and Image Classification

It is suggested that image enhancement should generally improves image classification performance regardless of the classification algorithms used. This is because image features are amplified by enhancing its contrast and resolution, thus providing more suitable input for further processes [5]. Therefore, the performance of image classification should be more accurate. Rodríguez-Rodríguez et al. [6] conducted a study to analyze the effect of image illumination enhancement techniques towards the classification performance using CNN. The findings show that some image enhancement algorithms have different behavior towards classification performance due to the nature of the dataset. To fill in the gap, Mitschke et al. [13] tried to combine multiple enhancement algorithms resulting in improved the classification performance in every dataset tested. Chen [14] investigated how various image enhancement algorithms influence the classification performance on both RGB and greyscale images. While he found that some algorithms performed better than original images using the greyscale dataset, some of them failed to improve the classification performance using the RGB dataset. The study also discovered that the same image enhancement algorithms can contrastingly influence performance when using different classification models. As dataset from various areas has different image properties, this paper will only explore the health sector.

D. Image Enhancement and Image Classification in Healthcare

In the treatment of internal organs diseases process, images obtained via X-ray or CT-scan are needed. By using these images, researchers have been developing many medical image classification methods to help in identifying the disease sooner [7], [8], [22]–[24], [12], [15]–[21]. But sometimes, the images obtained from either X-ray or CT-scan have poor visual images, such as low contrast on CXR images of tuberculosis patients [17], noise, low contrast, uneven illumination, and obscurity on retinal fundus images [18], low contrast and various types of noise on brain MRI images [21], etc. These images degradation can impact the performance and even loss of information in medical image classification [18]. Soomro et al. [15] suggested that enhancing images using CLAHE after being pre-processed using methods based on Fuzzy Logic and two datasets, give higher accuracy (95.9% and 96.1%) compared to most methods. Furthermore, Rahman et al. [7] suggested using five different image enhancements in the method resulting all models having better accuracy (96.29%) with enhanced images than the original one. Therefore, image enhancement techniques play a big part in improving medical image classification.

III. Methodology

We implemented the guidelines provided by Kitchenham, et al. [25] to systematically review the literature and report the results. A review protocol is imposed at the start of the study to reduce review biases. First, we define a set of research questions to fulfill the objectives. Then, the search strategy consisting of databases and string query is formulated to automatically find relevant publications. Afterwards, the resulting set of papers is systematically reviewed manually based on the selection criteria and exclusion criteria defined. Finally, the results are presented and discussed to answer the research questions.

A. Research Questions

Three research questions (RQs) were established to ease the process of gathering and analyzing essential information. The RQs are chosen to understand the image enhancement method and classification method used, and the effect of such approach. The review's analysis were based on these RQs.

RQ1: What are the problems and related datasets used in medical image classification?

RQ2: What are the methods of image enhancement related to medical image classification?

RQ3: Does image enhancement impact positively to the performance of CNN-Based deep learning for medical image classification?

B. Search Strategy

We executed the search on three internationally accepted databases: IEEE Xplore, Scopus, and ScienceDirect. Publication date restriction is also employed to easily find more recent publications. We formulated and executed the following search string on the forementioned databases to find the matching pattern in title, abstract, and keywords of the publications.

IEEEXplore: "Image Enhancement" AND ("Image Classification" OR "Medical Image Classification") AND ("CNN" OR "Deep Learning")

Scopus: "image" AND ("enhanc" OR "CLAHE") AND "classification" AND "medical" AND ("CNN" OR "Deep Learning")*

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ScienceDirect: "image" AND ("image enhancement" OR "CLAHE") AND "classification" AND "medical" AND ("CNN" OR "Deep Learning")

The search query consisted of three main parts separated by the term "AND". The first part is to ensure image enhancement is discussed in the publications. The second part is to find publications that classify medical image classifications. The more general keyword "Image Classification" is employed to help gather publications with no "medical" keyword mentioned (e.g., Tuberculosis Detection). The third part is used to ensure the paper used state-of-the-art CNN.

C. Study Selection Criteria

After acquiring publications using the defined search query, we filter out relevant studies based on the predetermined criteria. The selected publications must fulfill all selection criteria and none of the exclusion criteria shown in Table I.

TABLE I. SELECTION AND EXCLUSION CRITERIA

Selection Criteria	Exclusion Criteria
Internationally recognized academic paper and scopus indexed	Papers published prior to 2017 (especially for method)
Paper must cover image classification stages	Web documents
Paper must include image enhancement techniques	Papers in which the full text is unavailable
Paper must include comparison between the results of enhanced dataset and original dataset	-

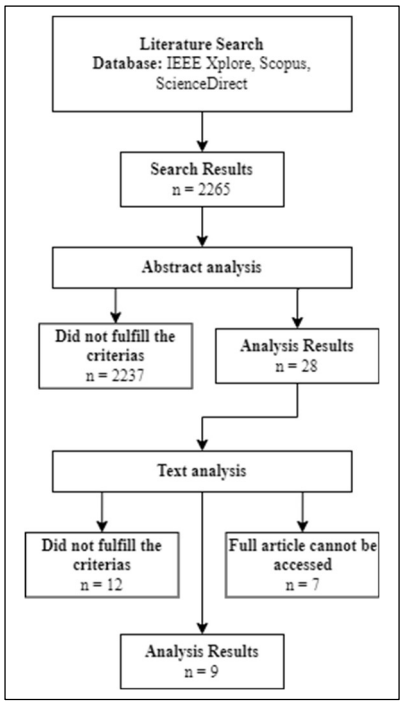

Fig. 1. Overview of literature identification process

The first screening of title and abstract eliminated 2237 papers as they did not relate to medical areas or the classification method is outdated (i.e., SVM, KNN, etc.). The remaining 28 papers were analyzed thoroughly. There was 12 paper that did not fulfill the criteria and 7 papers in which the authors could not access the full text. The selection produced 9 papers to be used for the analysis. The overview of the whole process can be seen in Fig. 1. Note that the ratio of selected papers compared to the initial search results is small due to most paper did not fulfill the fourth selection criteria.

IV. RESULTS AND DISCUSSION

A set of nine papers were selected for the detailed review. The overview of the papers is provided in the first subsection. Then, each paper is discussed based on the research question mentioned in the next subsections.

A. Overview of the Reviewed Papers

The reviewed papers comprised of one paper published in 2018, three papers published in 2019, three papers published in 2020, and two papers published in 2021. Most of the papers picked HE or its variations as one of the image enhancement methods. More explanation regarding image enhancement methods and CNN architectures used in each paper will be explained in section C and D respectively. In most cases, image enhancement methods managed to improve the overall classification performance.

B. RQ1: What are the problems and related datasets used in medical image classification?

Researchers have tried to detect various diseases using image classification. Table II describes the diseases and the related datasets used in each disease retrieved from the papers reviewed. Collectively, there are six different diseases used in the nine reviewed papers.

One of the diseases discussed in the reviewed papers is breast cancer [26], [27]. The examination is usually used on breast tissues to determine whether it is benign or malign. Hence, the classification agent is expected to separate tissues with malignant signs from normal ones using images in each dataset. Vidyarthi et al. [26] used 9,109 microscopic histological breast tumor images from 82 patients in the BreakHis dataset. Conversely, Lu et al. [27] used 9,927 mammogram images collected from 2,363 examinees of teaching hospitals in Taiwan.

Another disease tackled in some of the reviewed papers is diabetic retinopathy [12], [20]. The disease requires the agent to classify retinal fundus images in two classes: normal fundus and fundus with signs of diabetic retinopathy. There are three different datasets used in the related reviewed papers. Abu Samah et al. [12] used two public datasets, which are DIARETDB1 (89 color fundus images) and e-Ophtha (148 images with microaneurysms (MAs) or small hemorrhages and 47 images with exudates (EXs)). On the other hand, Hemanth et al. [20] used the MESSIDOR dataset for evaluation consisting of 400 retinal fundus images.

TABLE II. PROBLEMS AND RELATED DATASETS USED IN MEDICAL IMAGE CLASSIFICATION

Disease	Dataset	Ref
Breast Cancer	Breast Cancer Histopathological Image Classification (BreakHis) Dataset	[26]
	Dataset from Taiwan Teaching Hospital	[27]
Diabetic Retinopathy	DIARETDB1	[12]
	e-Ophtha	[12]
	MESSIDOR	[20]
Melanoma	International Skin Imaging Collaboration (ISIC) 2016 Dataset	[29]
Tuberculosis	Shenzhen Public Dataset	[17]
Pneumonia	Dataset from Guangzhou Women and Children's Medical Center	[30]

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COVID-19	BIMCV	[28]
	COVIDx	[28]
	COVID-CXNet	[28]
	COVQU	[7]

The disease with the most datasets used is COVID-19 [7], [28]. Every dataset consists of chext X-ray (CXR) images labeled normal or infected with the disease. In total, there are four datasets used exclusively to detect COVID-19. Qi et al. [28] used a total of 18,219 CXR images from three different datasets: BIMCV, COVIDx, and COVID-CXNet. In contrast, Rahman et al. [7] used 18,479 CXR images from the COVQU dataset.

C. RQ2: What are the methods of image enhancement related to medical image classification?

From the reviewed papers, various image enhancement techniques are used as the pre-processing step in the proposed models. Table III lists all image enhancement techniques used and their effect on images. In general, image enhancement methods are applied to complete one or more of these tasks: enhance the image contrast, increase the image sharpness, remove noises in the image, and correct color differences to create more uniform datasets.

For image contrast enhancement, the most preferred approach is CLAHE. Most of the papers used CLAHE because it performs adaptive histogram equalization over a limited small region with minimum risk of adding noise to images. Another frequently used method is HE [7], [12], [20], [29] due to its simplicity and low computational cost. Cherepkova and Hardeberg [29] also include three other different methods: Spatio-Temporal Retinex-inspired Envelope with Stochastic Sampling (STRESS) Contrast Enhancement, and Automatic Enhancement. Finally, Rahman et al. [7] used Balanced Contrast Enhancement Technique (BCET) to maintain the overall shape of the histogram, unlike CLAHE. Another enhancing technique is noise reduction. One of the methods used is the median filter (MF), where each pixel in an image is replaced with the median value of its neighboring pixels [27]. However, Wu et al. [30] stated that this approach often causes false noise detection. Therefore, they proposed adaptive median filtering by combining the normal MF with the mean filter for better image denoising results.

TABLE III. COMPARISON OF IMAGE ENHANCEMENT METHODS RELATED TO MEDICAL IMAGE CLASSIFICATION

Image Enhancement Method(s)	Effects to the image	Ref
CLAHE	Improves contrast and edges in a limited region	[7], [17], [20], [26], [27]
Median Filter	Removes noises while preserving edges in an image.	[27], [30]
HF	Sharpen image and significantly changes color intensity.	[12]
HE	Stretches dynamic range in the histogram	[7], [12], [20], [29]
Retinex	Improves dark regions inside an image and equalizes color tone	[29]
STRESS	Enhances contrast and applying a color correction to make images more uniform.	[29]
AWB	Changes color temperature of images based on their deviation from the neutral tone.	[29]
Contrast enhancement	Enhances hidden edges and structures of a blurry image.	[29]
Automatic enhancement	Increases contrast and correct image color.	[29]
UM	Produces a less blurred image and amplifies high-frequency regions in the image.	[17]
HEF	Emphasize and accentuate image edges and lowers image contrast.	[17]
Adaptive Median Filter	Removes noises in an image without removing unpolluted pixels.	[30]
Local Phase Enhancement	Creates a clearer separation between parts with low and high intensity in an image.	[28]
Image Complement	Inverts the value of pixels inside an image.	[7]
GC	Corrects pixel based on relation between pixel and gamma value	[7]
BCET	Modifies the contrast of an image while preserving the histogram pattern of the image.	[7]

Image sharpening is also used in several papers. Munadi et al. [17] used two different image sharpening algorithms, Unsharp Masking (UM) and High-frequency Emphasis Filtering (HEF). They stated that applying HEF to an image lowers its contrast; thus, they recommend using a contrast enhancement algorithm such as HE to increase sharpness and contrast. Abu Samah et al. [12] applied the HF that enhances high-intensity regions, but this method significantly impacts the image's color intensities. They chose HE to correct the image's color. Other methods that combine contrast enhancement with image sharpening are Contrast Enhancement and Automatic Enhancement [29]. Overall, it is suggested that contrast enhancement is used to counter the drawback caused by image sharpening.

The reviewed papers also used other methods than the previously mentioned approaches. Automatic White Balance (AWB) corrects the image based on its color temperature [29]. Gamma Correction (GC) changes pixel values based on the relationship with its gamma value [7]. Retinex [29] aims to enhance dark regions inside an image. Local Phase Enhancement [28] separates parts of an image with low intensity and high intensity. Lastly, Rahman et al. [7] used Image Complement, a method that inverts the whole image color, switching dark and light areas in an image.

D. RQ3: Does image enhancement impact positively to the performance of CNN-Based deep learning for medical image classification?

To observe the effectiveness of image enhancement techniques towards the classification performance, the classification model and the measured evaluation metrics using the original dataset and the enhanced dataset is compared as presented in Table IV.

All reviewed papers do not use an existing CNN-based model that simultaneously enhances and classifies the datasets. Instead, image enhancement is applied beforehand, and the model is purely used to perform classification. Six papers out of nine trained their own CNN model from scratch. Some CNN models are modified with other classifiers, namely RF and SVM. Meanwhile, four papers used pre-trained models such as AlexNet, ResNet, ChexNet, DenseNet, EfficientNet, Inception, SonoNet, and Xnet as their classification approach.

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Generally, there are two major metrics used to evaluate the classification: accuracy and F1-score. Accuracy is used when correct classification of the corresponding class is more important. Meanwhile, F1-score serves a better metric if the data is imbalanced. When comparing the performance between the original dataset and the enhanced dataset, image enhancement gives an overall improvement to the classification performance in both metrics. This is because the image enhancement algorithms can make hidden features from images more visible. Contrast enhancement algorithms such as CLAHE and HE have shown improvements in almost all papers due to how low the image contrast is in each dataset [7], [20], [26], [27], [30]. Furthermore, color enhancement algorithms such as Retinex and AWB performed better if the task requires color detection to separate malign images [29], while intensity enhancement algorithms such as Gamma Correction and Local Phase Enhancement are best suited when features are hidden in dark regions [7], [28]. In addition, sharpness enhancement algorithms such as HF and UM also contributes positively to the results when image details are crucial [17], although sometimes used with other enhancement methods such as contrast enhancement and color enhancement to counterbalance its disadvantage [12].

However, some image enhancement algorithms fail to boost the performance. Cherepkova and Hardeberg [29] show that Automatic Enhancement and Retinex perform worse than most image enhancement techniques used. Automatic Enhancement mostly failed in the full-size colored images. It increases contrast and corrects image color, but it does not contribute positively to the colored images. Meanwhile, Retinex positively impacts the classification on colored images, but performs worst on grayscale images. They demonstrate that using Retinex may overexpose the images. Due to this, features from greyscale images become more hidden when using Retinex than when using original images. Meanwhile, Munadi et al. [17] analyzed that enhancing images using CLAHE on CXR images will lead to worse performance than the original one after the transfer learning process. By using Lightness Order Error (LOE) measurement, CLAHE achieved 1852, while HEF achieve only 341. A lower LOE score indicates that the image is more natural. Due to this, the image enhanced by CLAHE became unnatural, resulting in worse performance.

TABLE IV. COMPARISON OF CLASSIFICATION PERFORMANCE WITH AND WITHOUT APPLYING IMAGE ENHANCEMENT TECHNIQUES

Ref	Image Enhancement Techniques	Classification Method	Performance without Image Enhancement	Performance with Image Enhancement
[26]	CLAHE	Deep Model LeNet-5	Accuracy: 86.24%	Accuracy: 90.39%
[27]	MF, CLAHE	Transfer Learning using Deep CNN	F1-score: 88.0%	F1-score: 88.3%
[12]	HF, HE	CNN using MATLAB 2017b platform	**E-Optha** (Acc: 97.3%), **DIARETDB1** (Acc: 92.8%).	**E-Optha** (Acc: 99.8%), **DIARETDB1** (Acc: 96.4%).
[29]	Retinex, STRESS, AWB, Contrast Enhancement, Automatic Enhancement, HE	CNN and SVM based MatConvNet function	**Full Size Image** (F1: 39%), **Grayscale Image** (F1: 41%), **Grayscale Reduced Size Image** (F1: 39%)	**Full Size Image** (Worst F1: AE 40%, Best F1: Retinex 43%), **Grayscale Image** (Worst F1: Retinex 36%, Best F1: CE & AE 41%), **Grayscale Reduced Size Image** (Worst F1: AE & HE 35%, Best F1: CE 41%)
[17]	UM, HEF, CLAHE	Transfer Learning	Average Validation Accuracy: 86.27%	Based on average Validation Accuracy between each model **HEF:** 88.51%, **UM**: 86.83%, **CLAHE**: 85.43%
[30]	Improved Adaptive Median Filter	CNN, CNN-SVM, CNN-RF, ACNN-SVM, ACNN-RF	**CNN** (Acc: 94.50%, F1: 94.18%), **CNN-SVM** (Acc: 96.27%, F1: 95.35%), **CNN-RF(**Acc: 95.84%, F1: 95.03%)	**ACNN-SVM** (Acc: 96.36%, F1: 97.26%) **ACNN-RF (**Acc: 96.85%, F1: 97.73%)
[20]	HE, CLAHE	CNN	Accuracy: 90.75% F-score: 81.50%	**HE** (Acc: 94.33%, F1: 88.67%), **CLAHE**(Acc: 97.00%, F1: 94.00%)
[28]	Local Phase Enhancement	Pre-trained models	**Data 1** (Acc: 94.58%) **Data 2** (Acc: 93.4%)	**Data 1** (Acc: 95.57%), **Data 2** (Acc: 94.44%)
[7]	HE, CLAHE, Image Complement, GC, BCET	Pre-trained models	Acc: 93.22% F1: 93.22%	**Image Complement** (Acc: 93.46%, F1: 93.47%), **HE** (Acc: 93.44%, F1: 93.42%), **CLAHE** (Acc: 93.90%, F1: 93.89%), **GC** (Acc: 95.11%, F1: 94.53%), **BCET** (Acc: 94.12%, F1: 94.14%)

V. CONCLUSION

This study has reviewed nine papers to examine the effects of image enhancement methods to the performance of medical classification models. The problems that were covered in those papers consists of various problems, including but not limited to breast cancer, melanoma, and COVID-19. Datasets varied between each disease. Furthermore, various image enhancement techniques were used, with CLAHE and HE as the most frequently used image enhancement algorithms.

In most situations, using image enhancement on the dataset beforehand can improve the classification performance of medical images. However, some algorithms fail to perform well than the original data in some circumstances. Nevertheless, the study has shown that image enhancement has great potential in improving different approaches of medical image classification issues. Because each medical dataset has different difficulties, it requires different types of image enhancement. In future works, we would like to investigate on whether image enhancement algorithms are needed to a specific medical dataset. In addition, an analysis of which image enhancement method

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works the best on a certain dataset is an interesting topic for future studies.

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